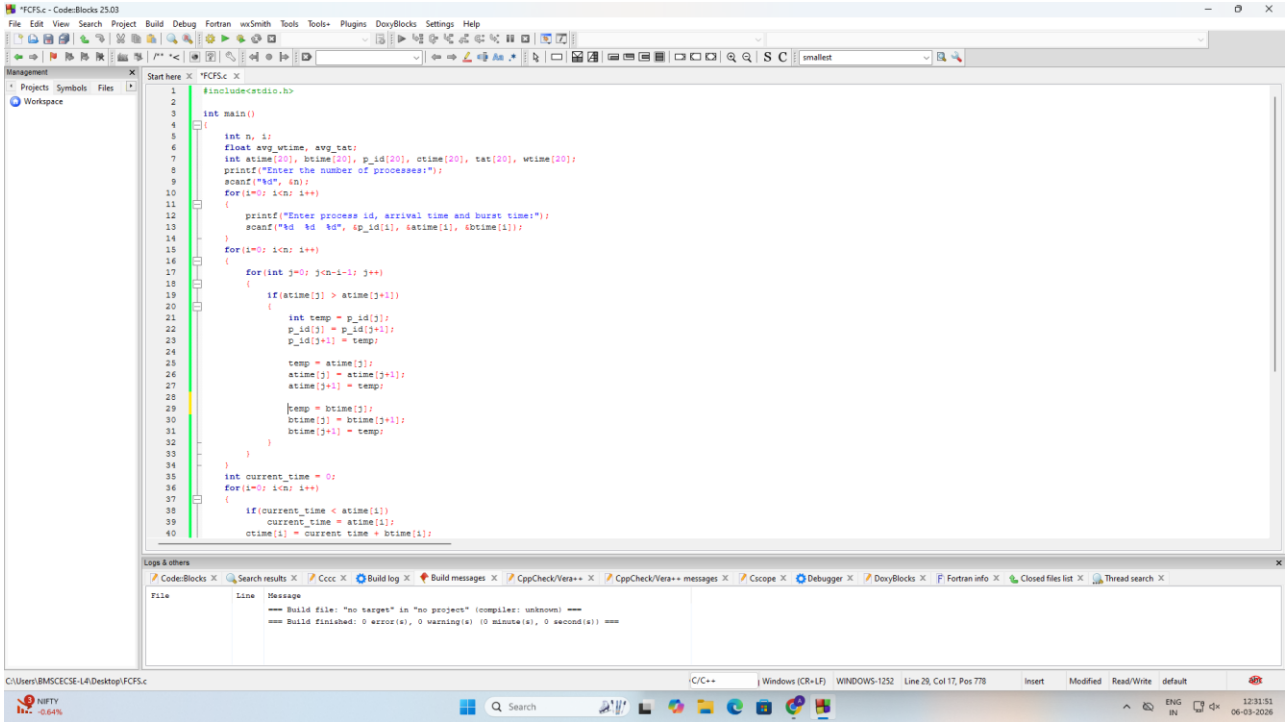
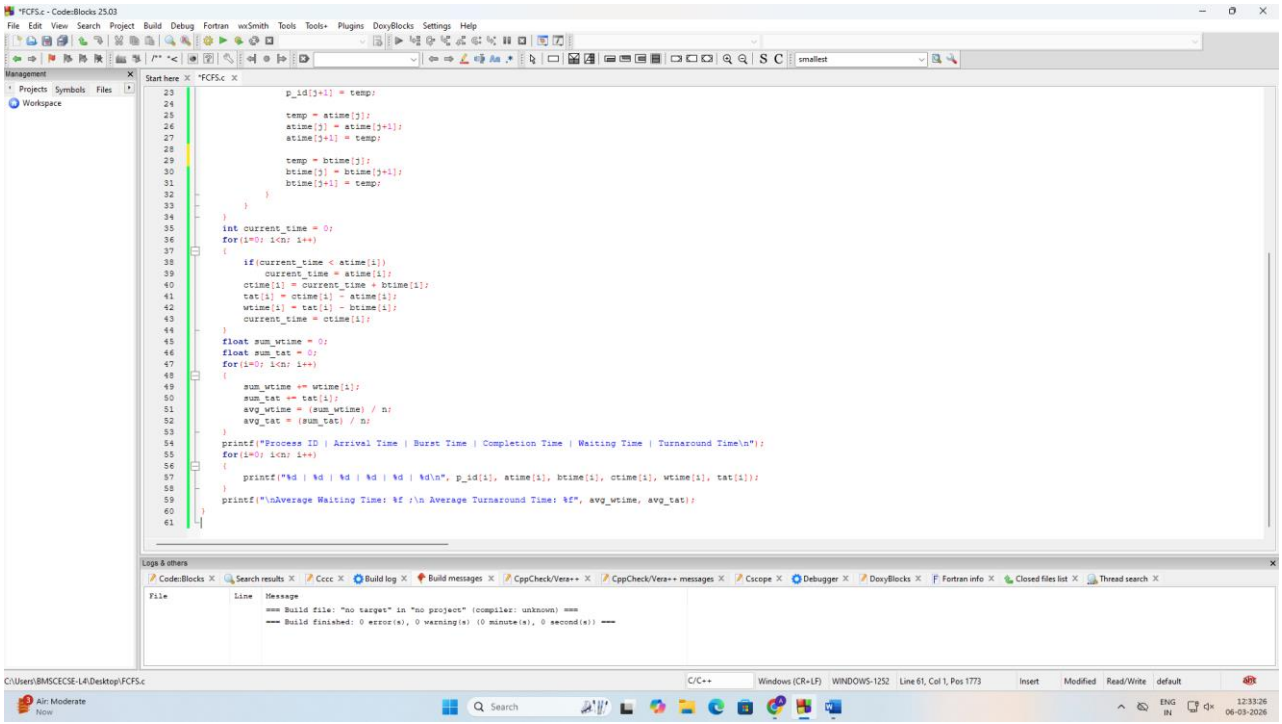


```c
#include<stdio.h>

int main()
{
    int n, i;
    float avg_wtime, avg_tat;
    int atime[20], btime[20], p_id[20], ctime[20], tat[20], wtime[20];
    printf("Enter the number of processes:");
    scanf("%d", &n);
    for(i=0; i<n; i++)
    {
        printf("Enter process id, arrival time and burst time:");
        scanf("%d  %d  %d", &p_id[i], &atime[i], &btime[i]);
    }
    for(i=0; i<n; i++)
    {
        for(int j=0; j<n-i-1; j++)
        {
            if(atime[j] > atime[j+1])
            {
                int temp = p_id[j];
                p_id[j] = p_id[j+1];
                p_id[j+1] = temp;

                temp = atime[j];
                atime[j] = atime[j+1];
                atime[j+1] = temp;

                temp = btime[j];
                btime[j] = btime[j+1];
                btime[j+1] = temp;
            }
        }
    }
    int current_time = 0;
    for(i=0; i<n; i++)
    {
        if(current_time < atime[i])
            current_time = atime[i];
        ctime[i] = current_time + btime[i];
```



```c
                p_id[j+1] = temp;

                temp = atime[j];
                atime[j] = atime[j+1];
                atime[j+1] = temp;

                temp = btime[j];
                btime[j] = btime[j+1];
                btime[j+1] = temp;
            }
        }
    }
    int current_time = 0;
    for(i=0; i<n; i++)
    {
        if(current_time < atime[i])
            current_time = atime[i];
        ctime[i] = current_time + btime[i];
        tat[i] = ctime[i] - atime[i];
        wtime[i] = tat[i] - btime[i];
        current_time = ctime[i];
    }
    float sum_wtime = 0;
    float sum_tat = 0;
    for(i=0; i<n; i++)
    {
        sum_wtime += wtime[i];
        sum_tat += tat[i];
        avg_wtime = (sum_wtime) / n;
        avg_tat = (sum_tat) / n;
    }
    printf("Process ID | Arrival Time | Burst Time | Completion Time | Waiting Time | Turnaround Time\n");
    for(i=0; i<n; i++)
    {
        printf("%d | %d | %d | %d | %d | %d\n", p_id[i], atime[i], btime[i], ctime[i], wtime[i], tat[i]);
    }
    printf("\nAverage Waiting Time: %f ;\n Average Turnaround Time: %f", avg_wtime, avg_tat);
}
```

Program 1: First Come, First Serve (FCFS)